

Concrete Mix Design(ACI Method)

Quality Control Laboratory, LGED-Rangpur

Ref.Memo No-46.02.8576.001.24-623

Date-14/10/2024

Remaining work of 294.00m Long PSC Girder Bridge on over Kartoa river at Joyntopur ghat, Gopinathpur road, at ch= 1500m

Upapazila-Pirganj, Dist- Rangpur. (for ~~Cross~~ Girder)

Contractor- AK-SB-MA(JV), Jummapara, Rangpur.

Package No-CIB-Ran/W-37

Date of Test- 23-10-2024

1 Stipulation for proportioning:

Concrete Grade/Class-(40Mpa)= C-40

Type of Cement Seven Rings Gold (OPC)

Minimum Cement content

Maximum nominal size of Aggregate

Maximum Water Cement Ratio

Slump

Admixture

Water reduction for Admixture

Sp Gr. Of Admixture

Aggregate:

408.000	kg/cm ² f_c =	40.0	Mpa
OPC			
400.000	Kg/cum		
20.000	mm		
0.450			
100-150	mm		
0.5%	cc/Bag cement		
10%			
1.16			

Aggregate	Max. Size mm	FM	Unit Weigt	Dry Bulk Sp Gr	Absorptn (%)	Present Moisture
CA	20		1776	2.725	0.718	0.2
FA		2.8		2.57	1.300	2.0

Mixing water(Chart or Trial)

Air Entrapped(Slump chart)

Targetted strength

W/C Ratio

Cement Content

Coarse Aggregate

30% more

160.000	kg/cum
2.000	%

530.400 kg/cm² f_{cr} = 52.01 Mpa

0.31

0.305

524.590 kg

524.590 kg

1160.438 kg

0.65 cum(Char)

2 New Absolute Volume:

Cement

Water

CA

Air

Admixture

FA

Wt of FA

Total

0.167	cum
0.160	cum
0.426	cum
0.020	cum
0.002	cum
0.775	cum
0.225	cum
579.158	kg

3 Batch Weight for 1 cum:

Cement

Water

CA

FA

Admixture

524.6 kg

160.0 kg

1160.4 kg

579.2 kg

2.623 kg

4 Ratio C : FA : CA by weight =

1.00	1.10	2.21
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